ROHIT PANWAR

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Software Engineer | Full-Stack & Machine Learning

EDUCATION

Bachelor of Computer Applications (BCA), Manipal University Jaipur — Expected 2027
12th (PCM), Senior Secondary School, Phalodi — Percentage: 74.6%

SKILLS

Languages:	Python, JavaScript (ES6+), C++, C#
Web & Mobile:	React, Next.js, React Native, Tailwind CSS, HTML, CSS
Backend:	Node.js, Express.js, FastAPI, Flask, REST APIs, WebSockets, JWT / NextAuth, Google OAuth
Machine Learning:	TensorFlow, 2D CNN, Scikit-learn, OpenCV, InsightFace (ArcFace), NLP, TF-IDF, Recommender Systems, Time-Series, Feature Engineering, Logistic Regression, Naive Bayes, Random Forest
Signal Processing:	Audio Signal Processing, FFT / STFT, Spectrograms, Voice Activity Detection
Data & Tools:	NumPy, Pandas, Streamlit, Git, GitHub, Postman, VS Code
Databases:	SQL, MongoDB, MongoDB Atlas, Mongoose
Deployment & Services:	Docker, VPS, Vercel, Render, Firebase Cloud Messaging, Google Maps, Cloudinary, Gemini API

PROJECTS

ZaikoMart — *React Native / Next.js / Google Maps / Firebase / Cloudinary*

- Built and launched a full e-commerce platform live in Phalodi, used by 1000+ users and 50+ local shops; customers order directly from shops through the website and the Android app.
- Built shop management so owners can fully manage and customize their own store: products, orders, delivery rules and store settings.
- Integrated Google Maps: users pick their location, nearby shops and products are shown based on pincode, and delivery charge is calculated from the distance between shop and customer.
- Developed a React Native Android app (published on Google Play) and a Next.js website that share one backend API.
- Integrated Google OAuth login, Firebase Cloud Messaging push notifications and Cloudinary image uploads.

Website • Play Store

FaceReg: Real-Time Face Recognition — *FastAPI / WebSockets / InsightFace / MongoDB / Next.js*

- Built a live multi-face recognition system: InsightFace (ArcFace) converts each face into a 512-dimensional embedding, matched by cosine similarity with a configurable threshold and confidence scores.
- Replaced per-frame REST calls with a WebSocket pipeline: frames stream to the backend and bounding boxes with match results stream back, supporting multiple faces per frame.
- Implemented client-side backpressure (send the next frame only after the previous response), so throughput adapts to inference speed with no queue buildup or overlay lag.
- Cached all embeddings in memory as one normalized NumPy matrix, refreshed only on registration, so matching is a single matrix multiplication with zero database calls during live recognition.
- Fit the model within a 512 MB free-tier host by switching to a lighter InsightFace variant and reducing detection resolution; deployed on Render and Vercel.

Live Demo • Frontend Repo • Backend Repo

Voice Command Recognition *TensorFlow / 2D CNN / FastAPI / Docker / Next.js*

• Built a voice command recognition system from scratch: audio converted to spectrograms using STFT (FFT) and classified by a 2D CNN; the browser records short clips and the FastAPI backend returns the predicted command.

• Found through per-class confusion matrices that a classical baseline was weak on rare words (22% to 89% recall, tracking sample counts); redesigned labels into Command / Unknown / Silence and balanced every class.

• Reduced CNN overfitting (97% train vs 92% validation, rising validation loss) with L2 regularization, SpatialDropout2D and SpecAugment-style audio augmentation.

• Built real-time voice activity detection with two-stage calibration (room noise and speaking volume), a pre-buffer for quiet word onsets and dynamic silence trimming.

• Verified the inference code against held-out test files (29/30 correct) to isolate an accent domain shift, then improved generalization with accent-diverse data from the Multilingual Spoken Words Corpus, capping additions per word to keep classes balanced.

• Deployed the FastAPI backend in Docker on Render and the Next.js frontend on Vercel.

Live Demo • Frontend Repo • Backend Repo

YouTube Clone *Next.js / Node.js / Express / MongoDB / Cloudinary*

• Built a full-stack YouTube-like application with a production-ready, reusable Express backend and a Next.js frontend.

• Implemented secure authentication (NextAuth credentials + JWT, bcrypt) with access and refresh token flows.

• Built REST APIs for users, videos, playlists, comments, likes, subscriptions, watch history and a channel dashboard with stats; wrote performant queries using Mongoose aggregation pipelines.

• Handled video, thumbnail and avatar uploads with multer and Cloudinary.

• Deployed the backend on Render and the frontend on Vercel; designed APIs with scalability and modularity in mind.

Frontend Demo • Frontend Repo • Backend Repo

SmartCatalog AI *Next.js / Gemini API / MongoDB*

• Built an AI commerce assistant that generates product categories, subcategories, SEO tags and sustainability tags as structured JSON using Gemini and stores results in MongoDB.

• Built a support chatbot with order status lookup from the database and refund/return escalation detection; logged AI prompts and conversations.

• Structured the system in layers (UI, API routes with input validation, AI service with prompt engineering, database) for maintainability.

Live Demo • GitHub Repository

Hybrid Movie Recommendation System *Machine Learning / Recommender Systems*

• Designed a hybrid recommender combining user-based collaborative filtering (mean-centered), item-based collaborative filtering and content-based TF-IDF filtering on genres, trained on MovieLens (9,742 movies, 100,836 ratings, 610 users).

• Ranked merged candidates using 60% model score, 20% popularity and 20% recency to improve relevance and diversity.

• Served recommendations through a Flask REST API (home page and movie page strategies, search, pagination) and built a Streamlit demo app.

GitHub Repository

Human Activity Recognition (HAR) *Machine Learning / Time Series*

• Built a classifier for sitting, standing, walking, running and stairs from smartphone accelerometer and gyroscope data recorded on an Android phone.

• Preprocessed signals: merged both sensors by timestamp, removed warm-up noise, trimmed start and end, and dropped invalid readings; used 2-second windows at about 50 Hz with 50% overlap.

• Engineered time-domain statistical features (mean, standard deviation, magnitude) per window; compared Logistic Regression with other models such as Random Forest.

• Evaluated with session-separated train/test splits to avoid leakage; final Logistic Regression model reached about 95% accuracy.

GitHub Repository

Spam Message Detection *Machine Learning / NLP*

- Built an NLP spam classifier for SMS/text messages using text preprocessing and TF-IDF vectorization.
- Trained Logistic Regression and Naive Bayes models; evaluated with precision, recall, F1-score and ROC-AUC.

TECHNICAL ACTIVITIES

- Solved data structures and algorithm problems on LeetCode and HackerRank to improve problem-solving skills.
- Explored machine learning through hands-on projects in computer vision, audio signal processing, NLP, recommender systems and time-series classification.
- Regularly read engineering blogs and documentation to stay updated with web and machine learning best practices.

LEADERSHIP & INITIATIVE

- Independently designed, built, and deployed end-to-end projects across backend, frontend, mobile and ML, using Docker, Render, Vercel and VPS hosting.
- Shipped ZaikoMart to real users in my city and took ownership of architecture, debugging and iterative improvements based on feedback and scalability needs.